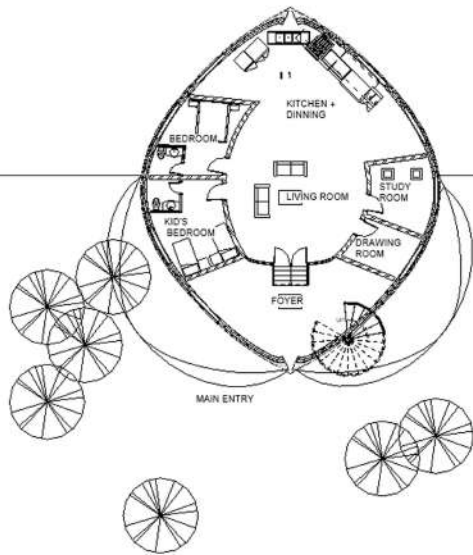
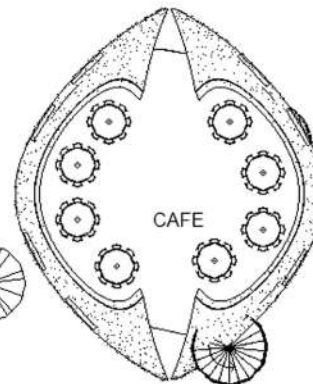


NAUTILUS

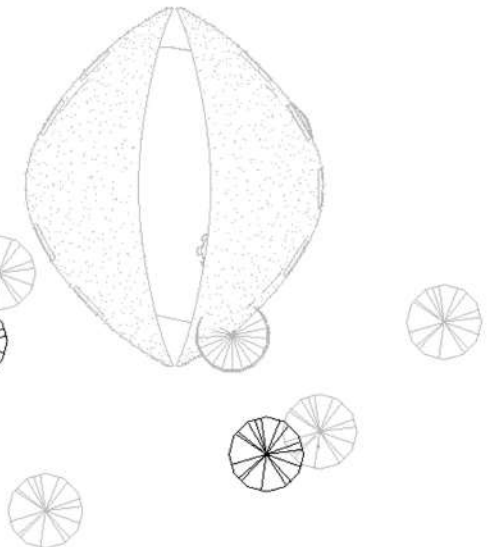
THE NAUTILUS WHERE SHORE BECOMES HOME



GROUND FLOOR PLAN



FIRST FLOOR PLAN



ROOF PLAN

SITE ANALYSIS

Location

Site: Krågevika, Lindesnes, Norway

Coordinates: 58°02'36.2"N, 7°09'18.5"E

Setting: Rocky coastal headland projecting into the North Sea

Project Type: Ocean bungalow partially submerged within the sea



Topography

The site naturally consists of:

Elevated rocky peninsula

Steep rock edges

Small protected bay

Natural depression between rock masses

Existing Site Conditions

Land

Hard granite rock formation

Minimal vegetation near shoreline

Small sandy beach (south-west)

Existing walking trail

Characteristics

Extremely high bearing capacity

Resistant to erosion

Excellent anchoring for marine foundations

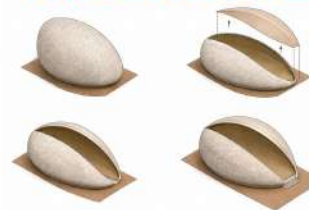
Geology

The coast is primarily:

Precambrian Granite

Gneiss Rock

FORM DEVELOPMENT



Base Mass: Solid conch-shell form.

Primary Opening: Slice and lift the upper shell.

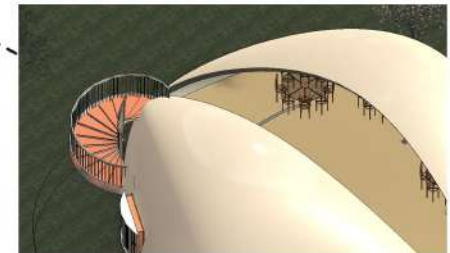
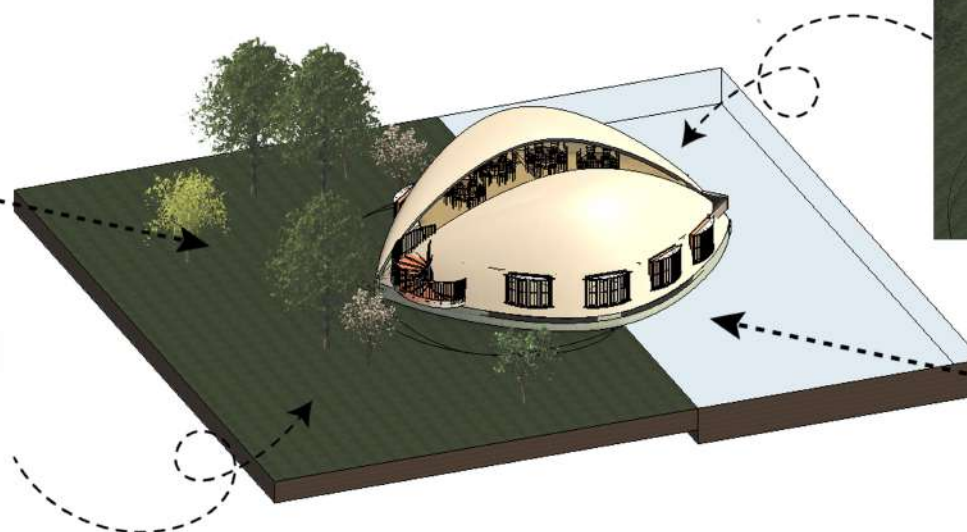
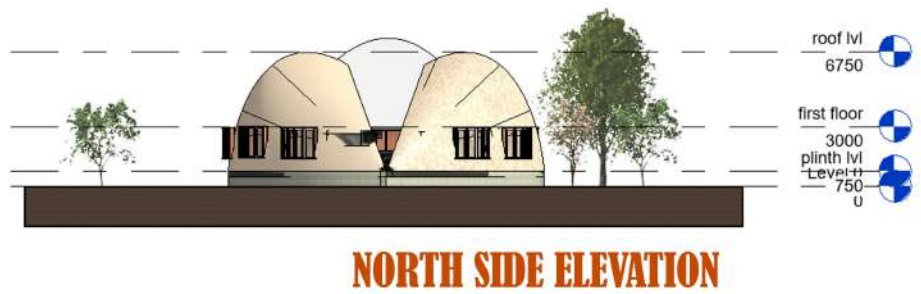
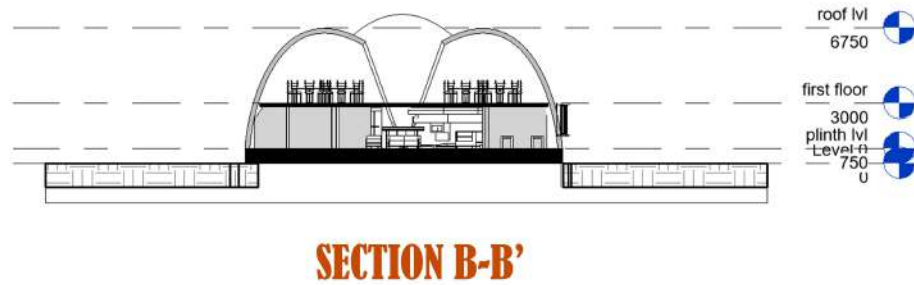
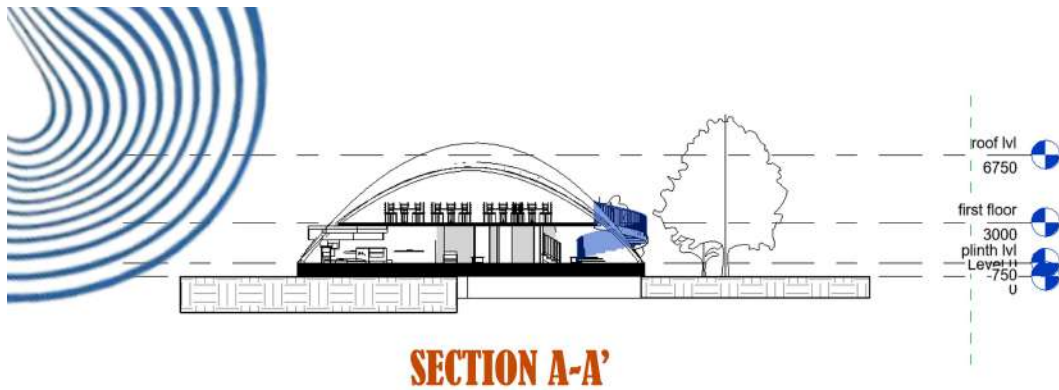
Movement: Show with dashed lines and arrows.

Carving: Hollow the interior, maintain shell thickness.

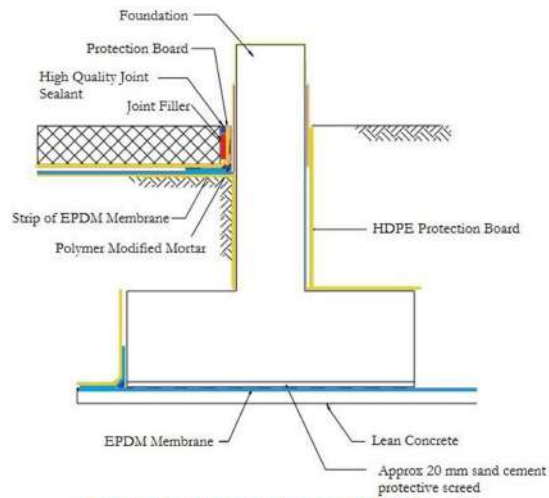
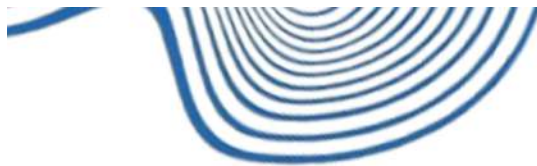
Entrance: Add front entrance cut.

Final Form: Raised upper shell, occupied lower shell.

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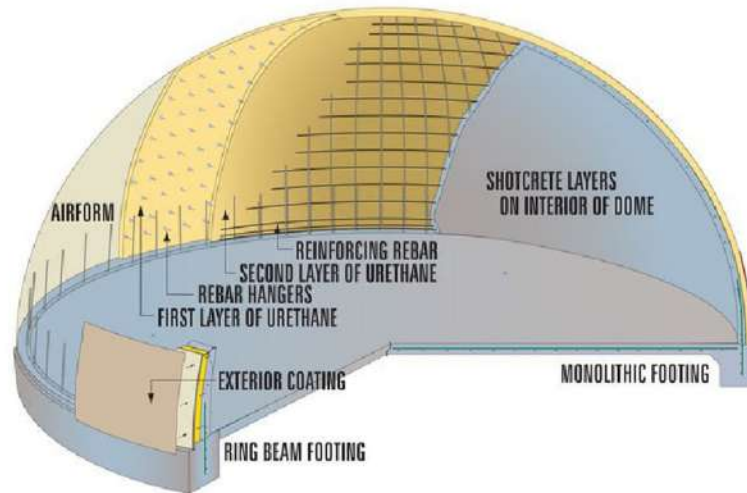
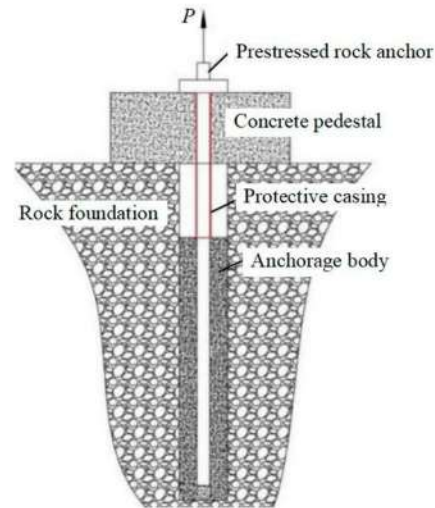


TUH-B-II-GRP-0184

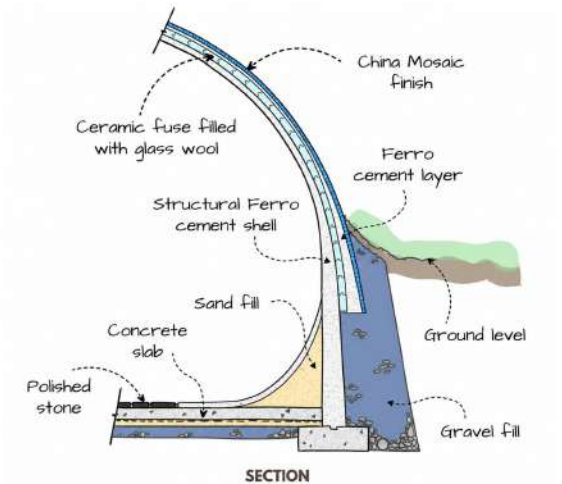


A rock-anchored foundation system is a type of foundation in which high-strength steel anchors are drilled and grouted into solid granite bedrock. The reinforced concrete foundation is then fixed to these anchors, transferring all structural loads directly into the rock.

Ideal for rocky coastal and underwater sites.
Provides excellent resistance to waves, wind, and uplift forces.
Prevents settlement by anchoring directly into strong granite.
Minimizes excavation and preserves the natural seabed.
Ensures long-term stability for marine structures like your conch shell ocean bungalow



For a marine structure inspired by a conch shell, ferrocement is an excellent choice because it creates smooth, seamless curved forms similar to shells.
Keeps the structure lightweight, reducing foundation loads in water.
Performs well against wave impact when properly reinforced.
Can be finished with marine-grade waterproof coatings for durability in saltwater.
Integrates well with steel or concrete substructures and marine piles.



Composition
Cement-sand mortar (typically 1:2 to 1:3 mix)
Galvanized steel wire mesh (woven or welded)
Small-diameter steel rods (6–10 mm) as a skeletal framework
Water with a low water-cement ratio for durability
Properties
Thin sections (20–50 mm thick)
High tensile strength due to wire mesh reinforcement
Excellent crack resistance
Lightweight compared to reinforced concrete
Can be moulded into complex curves and organic forms
Good impact and corrosion resistance (when properly protected)
Durable with low maintenance

